NATIONAL INSITITUTE OF TRANSPORT

DEPERTIMENT OF COMPUTING AND COMMUNICATION TECHNOLOGY

PROGRAM : HDIT

MODULE NAME : OPERATING SYSTEM

MODULE CODE : ITU 07309

LECTURE'S NAME : ALEXANDER MAKANTA

ACADEMIC YEAR : 2023/2024

CLASS : LEVEL 7 II

TASK : GROUP ASSIGNIMENT 02

GROUP NUMBER : 20

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SUBMISSION DATE :

TASK: Disk Operating System (DOS)/ WINDOWS and UNIX Commands for management files and permission.

The main commands for file management and permission handling in DOS/Windows and UNIX/Linux:

1. DOS / Windows Commands:

File Management

- **DIR**: Lists all files and folders in the current directory.
  DIR
- **COPY** source destination: Copies files from one location to another.
  COPY C:\example\file.txt D:\backup\file.txt
- DEL filename: Deletes a file.
  DEL file.txt
- REN oldname newname: Renames a file or directory.
  REN oldfile.txt newfile.txt
- MKDIR dirname or MD dirname: Creates a new directory.
  MKDIR newfolder
- RMDIR dirname or RD dirname: Removes an empty directory.
  RMDIR oldfolder
- MOVE source destination: Moves a file or folder.
  MOVE file.txt D:*newlocation*\

Permission Management (Windows Command Prompt and PowerShell)

1. **Command Prompt (ICACLS):**

- ICACLS filename /grant username:permissions: Grants specific permissions to a file.
  - ICACLS file.txt /grant John:F *# Full control for user Baraka*
- ICACLS filename /deny username:permissions: Denies specific permissions.
  - ICACLS file.txt /deny John:W *# Deny write permissions for Nuru*
- ICACLS filename /remove username: Removes permissions for a user.
  - ICACLS file.txt /*remove Baraka*

2. **PowerShell (ACL Commands):**

- Get Get-ACL path: Views permissions for a file or folder.
  -ACL C:*example\file.txt*
- Set-ACL path -AclObject $acl: Sets permissions using an ACL object.
  - $acl = Get-ACL C:*example\file.txt*
  - Set-ACL C:*example\file.txt -AclObject $acl*

2. UNIX / Linux Commands

File Management

- ls: Lists files and directories.
  ls -l *# Detailed list with permissions and ownership*
- cp source destination: Copies files.
  cp file.txt /*backup/file.txt*
- rm filename: Deletes a file.
  rm file.txt
- mv source destination: Moves or renames files.
  mv oldfile.*txt newfile.txt*
- mkdir dirname: Creates a new directory.
  mkdir newfolder
- rmdir dirname: Removes an empty directory.
  rmdir oldfolder

Permission Management

Permissions (chmod): Sets file permissions using symbolic or numeric notation.

- Symbolic:
  chmod u+x filename # Adds execute permission for the user
- Numeric:
  chmod 755 filename # Sets rwxr-xr-x permissions
- Ownership (chown): Changes the owner of a file.
  chown user:group filename
- Group Ownership (chgrp): Changes the group ownership of a file.
  chgrp groupname filename
- View Permissions:
  ls -l filename # Shows permissions, owner, and group